



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ASSIGNMENT 6

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Question 1

Implement a class Library that contains an array of Book objects.

Book class attributes:

- title
- author
- ISBN

Library class methods:

a. **bool addNewBook(title, author, ISBN)**

- Adds a new book to the list
- Return type: Boolean
- Use pass-by-reference
- Use scope resolution operator to access class members

b. **bool removeBooks(ISBN)**

- Removes book using ISBN
- Return type: Boolean
- Use pass-by-reference
- Define function outside the class

c. **displayDetails()**

- Display all books

Main function:

Add at least 5 books

Remove 1 book

Display list

```

1 #include <iostream>
2 using namespace std;
3
4 class Book {
5 public:
6     string title, author, ISBN;
7 };
8
9 class Library {
10     Book books[10];
11     int count;
12
13 public:
14     Library() { count = 0; }
15
16     bool addNewBook(string &t, string &a, string &i) {
17         if(count >= 10) return false;
18
19         books[count].title = t;
20         books[count].author = a;
21         books[count].ISBN = i;
22         count++;
23         return true;
24     }
25
26     bool removeBooks(string &i);
27
28     void displayDetails() {
29         for(int j = 0; j < count; j++) {
30             cout << books[j].title << " "
31                 << books[j].author << " "
32                 << books[j].ISBN << endl;
33         }
34     }

```

```

A X 1
B Y 2
D P 4
E Q 5

```

=== Code Execution Successful ===

```

37 bool Library::removeBooks(string &i) {
38     for(int j = 0; j < count; j++) {
39         if(books[j].ISBN == i) {
40             for(int k = j; k < count - 1; k++) {
41                 books[k] = books[k + 1];
42             }
43             count--;
44             return true;
45         }
46     }
47     return false;
48 }
49
50 int main() {
51     Library lib;
52     string t,a,i;
53
54     t="A"; a="X"; i="1"; lib.addNewBook(t,a,i);
55     t="B"; a="Y"; i="2"; lib.addNewBook(t,a,i);
56     t="C"; a="Z"; i="3"; lib.addNewBook(t,a,i);
57     t="D"; a="P"; i="4"; lib.addNewBook(t,a,i);
58     t="E"; a="Q"; i="5"; lib.addNewBook(t,a,i);
59
60     string rem = "3";
61     lib.removeBooks(rem);
62     lib.displayDetails();
63     return 0;
64 }

```

=== Code Execution Successful ===

Question 2

Modify Question 1 using constructors:

a. Add:

- Default constructor
- Parameterized constructor
- Copy constructor

b. Use this pointer instead of scope resolution

c. Implement **removeBooks(ISBN)**

d. Implement **displayDetails()**

e. Create array of objects using:

- Initializer list
- Dynamic initialization

```
1 #include <iostream>
2 using namespace std;
3
4 class Book {
5 public:
6     string title, author, ISBN;
7
8     Book() {
9         title = ""; author = ""; ISBN = "";
10    }
11
12    Book(string t, string a, string i) {
13        this->title = t;
14        this->author = a;
15        this->ISBN = i;
16    }
17
18    Book(const Book &b) {
19        this->title = b.title;
20        this->author = b.author;
21        this->ISBN = b.ISBN;
22    }
23 };
24
25 class Library {
26     Book *books;
27     int count;
28
29 public:
30     Library(int n) {
31         books = new Book[n];
32         count = n;
33     }
34 }
```

=== Code Execution Successful ===

```

34
35 ~   bool removeBooks(string i) {
36 ~       for(int j=0;j<count;j++) {
37 ~           if(books[j].ISBN == i) {
38 ~               for(int k=j;k<count-1;k++)
39 ~                   books[k] = books[k+1];
40 ~               count--;
41 ~               return true;
42 ~           }
43 ~       }
44 ~       return false;
45 ~   }
46
47 ~   void displayDetails() {
48 ~       for(int j=0;j<count;j++) {
49 ~           cout << books[j].title << " "
50 ~               << books[j].author << " "
51 ~               << books[j].ISBN << endl;
52 ~       }
53 ~   }
54 ~ };
55
56 ~ int main() {
57
58 ~     // Initializer list
59 ~     Book b1("A","X","1"), b2("B","Y","2");
60
61 ~     // Dynamic initialization
62 ~     Library lib(5);
63
64 ~     lib.displayDetails(); // empty initially
65
66 ~     return 0;
67 ~ }

```

=== Code Execution Successful ===

Question 3

Create a class Account with attributes:

- account number (const long)
- transaction ID (long)
- transaction type (string)
- balance (double)

Methods:

a. long depositAmount(to, from, amount)

- Returns transaction ID

b. long creditAmount(to, from, amount)

- Same as deposit

- Returns transaction ID

c. `void displayDetails() const`

- Display:
 - account number
 - balance
 - transaction history

d. Use const arguments in all functions

Main function:

- Create at least 5 accounts
- Demonstrate credit & debit operations

```

1  #include <iostream>
2  using namespace std;
3
4  class Account {
5      const long acc_no;
6      long transaction_id;
7      string type;
8      double balance;
9
10 public:
11     Account(long a, double b) : acc_no(a) {
12         balance = b;
13         transaction_id = 0;
14     }
15
16     long depositAmount(const long &to, const long
        &from, const double &amt) {
17         balance += amt;
18         type = "Deposit";
19         transaction_id++;
20         return transaction_id;
21     }
22
23     long creditAmount(const long &to, const long &from
        , const double &amt) {
24         balance -= amt;
25         type = "Debit";
26         transaction_id++;
27         return transaction_id;
28     }
29

```

Account: 101
Balance: 1300

=== Code Execution Successful ===

```

23-     long creditAmount(const long &to, const long &from
        , const double &amt) {
24         balance -= amt;
25         type = "Debit";
26         transaction_id++;
27         return transaction_id;
28     }
29
30-     void displayDetails() const {
31         cout << "Account: " << acc_no << endl;
32         cout << "Balance: " << balance << endl;
33     }
34 };
35
36- int main() {
37     Account a1(101,1000), a2(102,2000), a3(103,1500);
38
39     a1.depositAmount(101,0,500);
40     a1.creditAmount(101,0,200);
41
42     a1.displayDetails();
43
44     return 0;
45 }

```

Question 4) Write a program to add data objects of two different classes using friend functions.

```

1  #include <iostream>
2  using namespace std;
3
4  class B;
5
6- class A {
7     int x;
8 public:
9     A(int a) { x = a; }
10    friend int add(A, B);
11 };
12
13- class B {
14     int y;
15 public:
16     B(int b) { y = b; }
17     friend int add(A, B);
18 };
19
20- int add(A a, B b) {
21     return a.x + b.y;
22 }
23
24- int main() {
25     A a(5);
26     B b(10);
27
28     cout << "Sum: " << add(a,b);
29
30     return 0;
31 }

```

Sum: 15
 === Code Execution Successful ===
 ENINGS [Apply](#)

Question 5

Create a class Complex with:

- Real part
- Imaginary part

Methods:

a. Parameterized constructor and copy constructor

b. `void display()`

- Display complex number

c. `void sum(Complex, Complex)`

- Add two complex numbers
- Implement as a friend function

```
1 #include <iostream>
2 using namespace std;
3 class Complex {
4     int real, imag;
5 public:
6     Complex(int r=0, int i=0) {
7         real = r;
8         imag = i;
9     }
10    Complex(const Complex &c) {
11        real = c.real;
12        imag = c.imag;
13    }
14    void display() {
15        cout << real << " + " << imag << "i" << endl;
16    }
17    friend Complex sum(Complex, Complex);
18 };
19
20 Complex sum(Complex c1, Complex c2) {
21     Complex temp;
22     temp.real = c1.real + c2.real;
23     temp.imag = c1.imag + c2.imag;
24     return temp;
25 }
26 int main() {
27     Complex c1(2,3), c2(4,5), c3;
28     c3 = sum(c1,c2);
29     c3.display();
30     return 0;}
```

6 + 8i

=== Code Execution Successful ===